

An FPGA-based Unidirectional Gateway Proposal for OT-IT Network Separation to Secure Industrial Automation Systems

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Abstract—A new FPGA-based approach for a data diode transmitting data unidirectionally from a high security zone to a lower security zone is evaluated and implemented. With the FPGA implementation, the need of additional hardware could be minimized compared to most state-of-the-art data diode realizations. The proposed data diode is designed to use the available backplane bus communication protocol between the programmable logic controller (PLC) and its connected peripherals. To demonstrate the universal approach an open-source platform based on the Revolution Pi (RevPi) and its backplane bus is used. Principally the data diode behaves as a mirror presenting the complete protected system behind the high security zone including the RevPi controller and its devices to the information technology (IT) network. Another controller within the lower security zone is used to imitate the operational technology (OT) process. Our approach is not limited to the above controller and can be applied to any type of PLC. The only requirement is that the communication protocol on the backplane bus is known and can be modified for this purpose.

Index Terms—Data Diode, FPGA-based, Unidirectional, Secure Industrial Automation System, OT-IT Separation.

I. INTRODUCTION

The availability of information from industrial assets and facilities to edge and cloud services create valuable opportunities for condition monitoring, metering or e.g. smart enterprise resource planning applications. At the same time, risk increases regarding the attack surface of the OT network, which remains challenging in today's applications [1]. Therefore, sensor, actuator and machine data being generated on the industrial shop-floor (OT network) and communicated towards the IT network must be isolated and separated for security reasons [2]. The proposed concept of this paper is an extension of the work-in-process paper presented in [3], also employing an open source approach [4], but with lower hardware and software requirements needed for the data diode. Here, the general approach is exemplified using the backplane bus

communication protocol of the RevPi [5] automation system to realize a prototype version of the data diode.

In IEC 62443, “Zones and Conduits” are stated to separate OT and IT networks into different security areas being connected by conduits. A defense in depth approach is implemented by employing firewalls as security measures to the conduits and by connecting areas with different levels of security within the OT network [6]. This is, e.g. especially essential for safety-critical applications [7], because the safety assessment changes and needs to be renewed if access possibilities to the system changes as well. For data diodes, also known as unidirectional (security) gateways, the inner network remains separated from the outer network without a signal path leading to the protected direction [8]. In this case, a new safety assessment of the entire system is not necessary.

This paper is organized as follows. Section II refers to the relevant related work. The general approach is described in Section III. In Section IV the communication protocol provided by the RevPi module is briefly reviewed, which is used by the data diode to collect and exchange data with the controller. Section V is focused in detail on the design of the data diode. In Section VI the testbench and measurement results are presented. The work is summarized in Section VII and the paper ends with an outlook on our future research.

II. RELATED WORK

A first scientific work on data diodes took place in the late 1980s. Fred Cohen [9] published the “digital diode” in 1988 to control the flow of information in IT networks. Okhravi and Sheldon first proposed the use of data diodes in industrial control systems in 2010 [10]. Data diodes are used successfully in the field for a long time and several vendors are established in the market. Joen and Na [11] list the main commercial products, all of which are based on Ethernet communication providing bandwidths of up to 10 Gbps. They support different but fixed protocols and services, such as file transfer, UDP communication or database replication, which are installed and configured individually. These commercial data diodes are designed to interconnect large site- or enterprise-wide OT and IT networks that require

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<https://dtecbw.de/home/forschung/hsu/digitalisierung-elektromobilitaet/>

customization when security features are added to the system or network. Larkin, et al. [12] acknowledge the usefulness and necessity of data diodes, but criticize the high financial cost and its limits to access large systems. To overcome the aforementioned barriers for a widespread use, we exemplify our new approach employing the open platform RevPi [5] automation system for secure unidirectional data exchange over an FPGA-based data path for smaller plant components or Internet of Things applications.

III. DESCRIPTION OF THE APPROACH

In the following, a data diode and a PLC, which we refer to as PLC-Mirror (PLCM), mirroring an industrial automation system behind the data diode into the IT zone is described. It is very beneficial that the structure of the entire system does not change from the point of view of the control system (e.g. SCADA) when adding a data diode. Nevertheless, the automation system is absolutely protected by the data diode and only the mirrored system can be accessed from the IT zone. Fig. 1 exemplarily illustrates the data diode integrated into an existing industrial automation system using RevPi modules employing a FPGA board [13]. The main idea is that the data diode monitors process data from the devices connected to the PLC in the protected area (hereinafter referred to as PLC), transmits this data unidirectionally and writes it into the process image of the PLCM. In this way, the PLC communication with its connected devices is mirrored, i.e. emulated, by the data diode. Since the data is only transported unidirectionally, the PLC and its devices are completely protected and unaffected in the event of a cyberattack. The data diode can easily be integrated into an already existing PLC system architecture. For the collection or exchange of process data between the PLC and the data diode or between the data diode and the PLCM, respectively, the PLC backplane bus is used. In Section VI, an experimental setup is realized as shown in Fig. 1 employing RevPi modules. For this reason, a detailed analysis of the communication protocol on the backplane bus

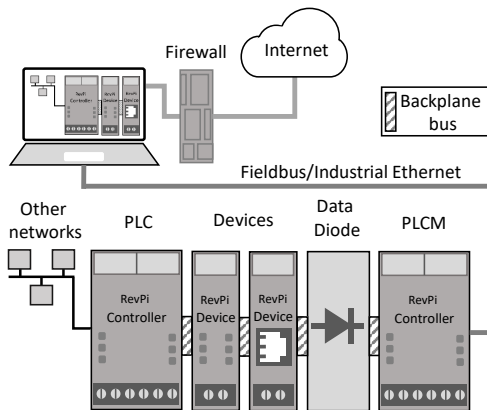


Fig. 1. An example of the data diode implementation in a simplified industrial automation system.

of the RevPi modules is analyzed and will be presented in the next section.

IV. COMMUNICATION PROTOCOL ON THE REVPI BACKPLANE BUS

The RevPi is an open source industrial PC (IPC) [5] with a modular architecture. The product family consists of three main groups: base modules serving as controllers, I/O expansion modules and fieldbus gateway modules serving as devices. With these devices and a built-in web application called PiCtory, an automation system can be easily built and configured according to customer needs. In order to connect several modules, the RevPi backplane bus, called PiBridge [14], is implemented on the back of each module. In addition to the ground pins, the PiBridge consists of three channels: the Position, Ethernet and RS-485 channel. These channels are used to transfer data between modules in various phases including: initialization, configuration, and cyclic data exchange. Fig. 2 shows an example of these communication phases for a RevPi controller, an I/O and a gateway device. The configuration and cyclic data exchange phase are performed on the RS-485 channel for an I/O device and on the Ethernet channel for a gateway device.

When the devices are connected to a controller module and are powered-up, the initialization phase begins. By reading the pin status of the position channel as low or high and configuring the pins as input or output, the devices know their corresponding position in the spatial sequence relative to the controller and the temporal sequence of initialization.

The initialization phase is performed from the innermost to the outermost device, first on the right side and then on the left side with respect to the controller. The initialization phase of each device is divided into other small processes; the exchange of device information and address allocation. During this phase the controller sends request packets with the appropriate command code and waits for the response packets from the device to be initialized. At the end of these phases, the information about the devices, such as serial number, device type number, etc., located in the "information of device"

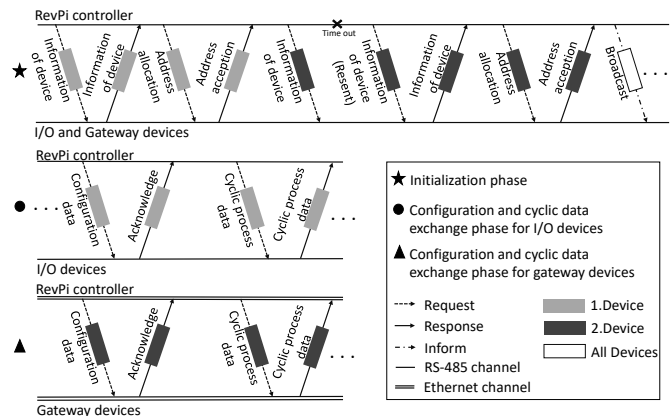


Fig. 2. Dataflow on the backplane bus for a RevPi controller and two devices. One device is an I/O device, whereas another device is a gateway.

response packets, is received by the controller, and each device is assigned with a unique address located in the "address allocation" request packets. If the controller does not receive a device response after a request command, an additional request command is sent after a predefined time. If the number of retries exceeds a predefined value, the controller considers the device to be faulty or not to be present.

The devices recognize the end of the initialization phase by means of a broadcast packet sent by the controller. Then, the configuration phase is performed for the I/O devices via the RS-485 channel and for gateway devices via the Ethernet channel. In this configuration phase, the configuration data that the user has set for the devices in the PiCtory application is transferred to the devices by the controller. Next, the devices adjust themselves according to the received configuration data.

At the end of the configuration phase, the successfully initialized and configured devices enter the cyclic data exchange phase on the corresponding channel of the backplane bus. Each cycle between the controller and an I/O device contains a request and a response packet lasting about 2.3 ms to nearly 5 ms, corresponding to a payload length between 8 bytes and 32 bytes depending on the configuration of the I/O device. The cycle time between gateway and controller is about 3 ms by default, corresponding to a total of 1024 bytes payload, i.e. 512 bytes for a request or response packet, respectively.

V. DATA DIODE DESIGN

The main task of the data diode is to collect data from the backplane bus of the PLC and to transmit these data unidirectionally to the PLCM via the other backplane bus. The backplane bus of the OT system is fully isolated from the backplane bus of the IT system. In this chapter the hardware and software architecture of the data diode is described in detail.

A. Hardware Design

A Xilinx's 7-series FPGA [13] has been used as hardware platform. For a better understanding a slightly simplified version of the hardware design is illustrated in Fig. 3. The data diode essentially consists of two separate zones, a high security zone and the low security zone, respectively. Each zone can only communicate with its assigned controller, the

PLC or the PLCM. Within each zone additional soft processors are implemented to support the individual applications, which are data collection and data processing. The only connection between these two zones is a group of modified dual-port block RAMs (MDBRAM). The MDBRAM builds on the dual-port block RAM (BRAM) [15] of the FPGA board. However, BRAM is modified so that in the low-security zone, writing into the BRAM is not supported anymore, i.e. the BRAM can only be accessed in a read-only mode by soft processor 3.

Fig. 4 shows the MDBRAM used in this work. Port B is a standard port with the following pins: memory address for reading or writing (addr), clock input (clk), data input (din) for writing data to the BRAM, data output (dout) for reading data from the BRAM, port enable (en), reset memory output latch or output register (rst), write enable (we) [15]. Port A also has similar pins. However, the pins for data input and write enable are no longer present. Thus, the read-only feature is implemented via port A. In Fig. 3 the soft processors 1 and 2 are connected to a standard port A being able to read data from or to write data into the MDBRAM. The soft processor 3, on the other hand, is connected to a read-only port B so that it can only read data out of the MDBRAM. With this method, data is transferred unidirectionally from application 1 and 2 to application 3, respectively, from the high security zone to the low security zone with the guaranteed security principle of the data diode.

To communicate with the Position, RS-485 and Ethernet channels of the backplane bus, the corresponding intellectual property cores (IP cores) provided by the Xilinx library and appropriate peripheral module (Pmod) interface are used as protocol converters for the soft processor. In the high security zone, soft processor 1 and its application are responsible for data collecting from the RS-485 channel, while soft processor 2 and its application collect data from the Ethernet channel. The Position channel is monitored by application 1 to detect the start of the PLC initialization phase in the OT network. The purpose of using two separate soft processors for two applications is to avoid the design of a complex interrupt service routine or multi-threading programming because soft processors have lower speeds and performance characteristics than hard processors.

The collected data is written to the MDBRAM and then

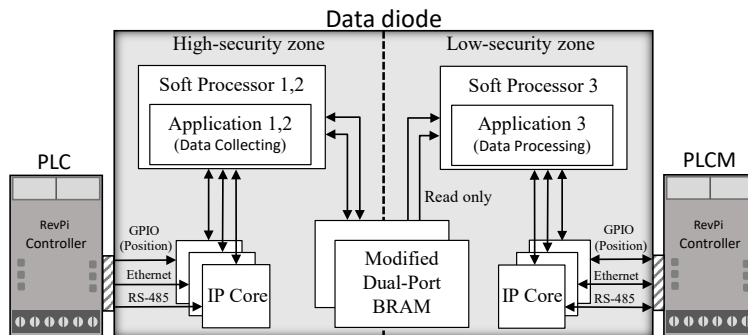


Fig. 3. The simplified structure of the data diode.

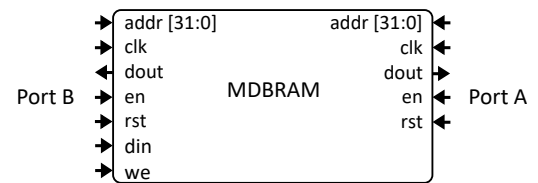


Fig. 4. Schematic of a modified dual-port BRAM.

read by soft processor 3 via the read-only port. This approach is considered as unidirectional data transport through the MDBRAM. Application 3 processes and sends the read-out of the MDBRAM data to the PLCM via the corresponding backplane bus.

B. Software Design

The main goals of the software design are the easy usability without major changes to the existing PLC configuration, the highest possible data rate on the backplane bus, and the lowest possible latency for unidirectional data transmission from the OT network to the IT network. Therefore, two different software modules are described, which were developed for two different modes of operation.

Mode 1: In this mode, application 1 and application 2, see Fig. 3, monitor the data exchange on the backplane bus of the PLC to collect information of devices, configuration data as well as process data from connected devices. The collected data is stored in predefined memory areas in the MDBRAM. By using the RevPi communication protocol, the data diode emulates a gateway device on the PLCM side. Thus, the PLCM perceives the data diode as a standard gateway device, exchanging data with the data diode over the backplane bus Ethernet channel. In the cyclic data exchange phase between the data diode and the PLCM, application 3 reads the necessary data from the MDBRAM and writes the data into an Ethernet packet with a payload of 512 bytes. Then, the data diode sends this packet to the PLCM, which replies with a response packet to confirm the transmission.

The advantage of this approach is that the data diode is invisible to the PLCM, but behaves like a normal gateway device. Therefore, the deployment of the data diode is straightforward, since the source code of the PLCM does not need to be modified to support the data diode. On the PLC side, the data diode sniffs the backplane bus, which has no impact on the architecture of the protected system. However, the total length of the data to be transferred unidirectionally should be less than 512 bytes. Therefore, some functions or the number of real I/O devices connected to the PLC may be limited, depending on the specific PLC settings, presuming that all process data shall be transferred from the PLC side to the PLCM side. If a real gateway device is connected with the PLC the amount of process data increases significant. Therefore, the user has to select the data, which shall be transferred. This conclusion shows that the data transmission rate needs to be increased, resulting in an improved mode 2 of operation.

Mode 2: The main idea of mode 2 is to modify the source code running on the PLCM by rewriting the driver for the RevPi backplane bus to increase the data transmission rate between the PLCM and the data diode. When using mode 2, the user configures the same architecture for the PLC and the PLCM system using PiCtory. As a result, the PLCM can reserve memory address space in the process image that fully corresponds to the address space assigned by the PLC for the connected devices. This is the first step in implementing the data diode as an architecture mirror. When the PLCM

TABLE I
SUMMARY OF DIFFERENCES BETWEEN TWO
SOFTWARE MODES IN SOME CRITERIA.

Criteria	Expected	Mode	
		1	2
Reflecting the PLC system	yes	no	yes
Source code of the PLCM changed	no	no	yes
Be able to transmit all process data	yes	no	yes

is powered-up and initialized, the data diode uses the RS-485 channel and the information of devices stored in the MDBRAM to replicate the initialization phase of the devices connected to the PLC within the PLCM. In the next phase, the data diode reads the configuration data received from the PLCM, compares these data with the corresponding data already received from the PLC and confirms to the PLCM if there is no error in the PLCM configuration so that finally both configurations are identical. Otherwise, the data diode informs the user to check the settings in PiCtory. In this way, the information and configuration of all devices connected to the PLC are written into the process image of the PLCM via the RS-485 channel. In other words, the devices connected to the PLC are emulated by the data diode on the PLCM side. During the cyclic data exchange phase on both sides of the data diode, process data stored in the MDBRAM are cyclically read by application 3 and mapped into an IP packet with a variable length N , which is forwarded to the PLCM via the Ethernet channel of the backplane bus. Using the modified driver [16], the PLCM first suspends to update process data from the RS-485 channel. Instead, the PLCM reads the process data from the received N -byte packet and writes them directly into the corresponding memory space reserved by the PLCM for the emulated devices.

This approach transfers all process data from the PLC side to the PLCM side with a time delay, which will be considered in the next section. Thus, the function or the number of I/O devices as well as the use of a gateway device for the PLC is no longer limited. The data diode acts like a mirror reflecting the entire system from the high security zone into the low security zone. So the structure of the entire system does not change from the point of view of an already installed control system (e.g. SCADA) after the integration of the data diode. Only the modified driver needs to be installed on the PLCM before mode 2 is able to operate. Tests showed that the normal operation of the PLCM is not affected by the modified driver. Table I summarizes the differences in key criteria for both modes.

VI. TESTBENCH

In this section, the data exchange across the MDBRAM as well as the smooth hard- and software operation is tested.

A. MDBRAM Verification

A regular dual-port BRAM has two ports with features such as reading data, writing data, resetting the output register, enabling or disabling the port. Within the MDBRAM one port

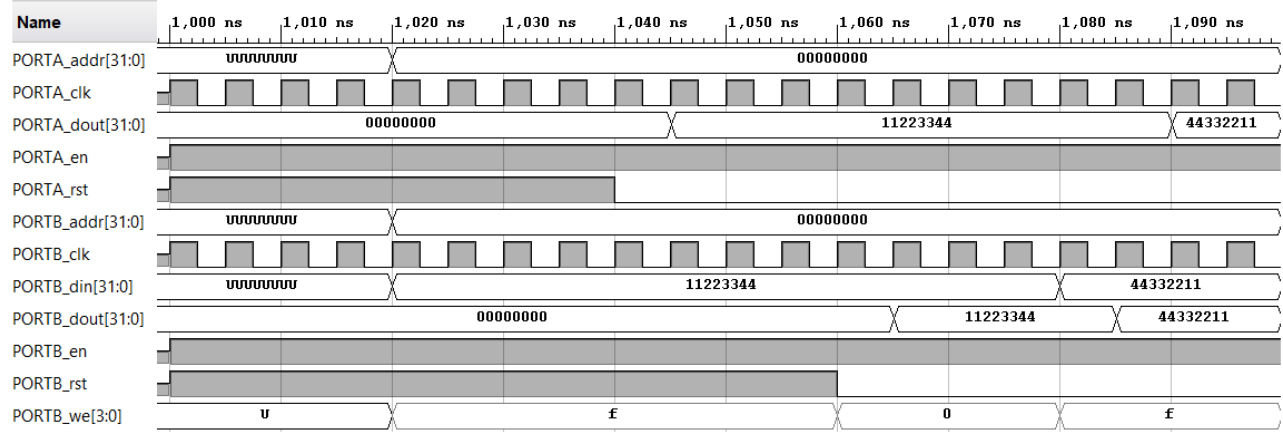


Fig. 5. Verify the MDBRAM feature through simulation using the Vivado 2019.1 [17].

is no longer able to write data while the other port supports the standard read/write functions. The verification process has been carried out running a behavior simulation at a clock of 200 MHz implemented in the Vivado application. The results of the simulation are shown in Fig. 5.

At 1,000 ns the clock input, the port enable and the reset pin are activated for both ports. In addition, no memory addresses are provided and the data output is set to 0x0000000 with a length of four bytes. After four clock cycles, at 1,020 ns, the memory address on both ports is set to 0x00000000 corresponding to the length of four bytes. At the same time the data input at port B is provided with a bit pattern to 0x11223344 and the write enable pin of port B is set to 0xf, equivalent to 0b1111. This means that the data input is written to memory at address 0x00000000 via port B. Since the reset pin on both ports is in the high status, the data output is still set to 0x00000000. At 1,040 ns the reset pin of port A is set to low status. After one clock cycle, the data output of port A is updated to 0x11223344. This results in port A being able to read the data port B stored at address 0x00000000. Similar to port B, one cycle after 1,060 ns, when the reset pin is switched to low status, the data stored at address 0x00000000 is read and appears at the data output of port B. At 1,080 ns, port B writes 0x44332211 into the memory with address 0x00000000. One clock cycle later, this data appears at the data output of port B. And after another cycle, the data appears at the data output of port A, resulting in at least two clock cycles for a four bytes data value to be written from port B and read at port A. In the actual testbench, a clock of 100 MHz is set for the MDBRAM to meet the timing requirements of the complexity of the total design. This means that the theoretically achievable maximum

throughput of the MDBRAM at this clock is 1.6 Gb/s. Another criteria often considered when designing multi-port BRAMs is the write-write race condition, which is not the case here since only one port is able to write data to the memory.

Therefore, the read, write, reset and port enable function of the two ports are verified. Since port A has no write enable and data input pin, it is not possible to write data to the memory cell in the MDBRAM via port A, thus the read-only feature of the MDBRAM has also been tested.

B. Experimental Data diode Test

In this subsection the usage of the data diode as illustrated in Fig. 1 is performed. The testbed consists of two RevPi Connect [5] serving as PLC and PLCM and one RevPi Digital Input Output (DIO) as I/O device, respectively. A simple industrial automation system is built using the above PLC and I/O device. A Zybo Z7 board [13] with an integrated FPGA has been chosen as data diode platform. The load factor of the hardware resources including Lookup Table (LUT), Flip-Flops (FF), BRAM, Input-Output pins (IO) is shown in Table II. After implementing the data diode, the FPGA chip has still available resources allowing additional design improvements. Three MDBRAMs with 8 KB each are used. The total amount of BRAM initialized for this design is larger than that actually needed and may be reduced in further optimization steps.

Table III shows the measured delay of the data transmitted through the data diode, which is essentially caused by

- the appearance of a packet to be read on the backplane bus of the PLC until this packet is written to the MDBRAM (PLC to MDBRAM),
- the existing of the written packet in MDBRAM until it is read from the other side of the data diode (Exist in MDBRAM),
- the reading of a packet in the MDBRAM until the successful writing of this packet into the process image of the PLCM (MDBRAM to PLCM).

In both modes, the relevant data is stored in the MDBRAM for the same amount of time depending on the cycle time on the Ethernet channel of the backplane bus between the data

TABLE II
HARDWARE UTILIZATION RESULTS OF THE DATA DIODE*.

Factor	LUT	FF	BRAM [KB]	IO
Available	53,200	106,400	630	122
Utilized	7,171(13.5%)	7,007(6.6%)	468(74.3%)	22(18%)

*The test is performed using the implementation of Vivado [17].

TABLE III

DELAY RESULTS OF THE DATA DIODE FOR THE TEST CONSISTING OF TWO REVPI CONNECT AS THE PLC AND THE PLCM AND A REVPI DIO AS I/O DEVICE.

Mode	PLC to MDBRAM ^a [Channel]: [ms]	Exist in MDBRAM [ms]	MDBRAM to PLCM ^a [ms]	Total delay [ms]
1	RS-485: 0.054 Ethernet: 1.281	0 - 3.000	4.196	3.250 - 7.250 5.477 - 8.477
2	RS-485: 0.054 Ethernet: 1.281	0 - 3.000	4.143 ^b	4.194 - 7.194 5.424 - 8.424

^aAll values are approximated and rounded based on 10 measurements with the PicoScope 5000 series [18] at the sampling rate of 1 GS/s.

^bThe packet length is $N = 512$, so the delay of the two modes can be compared under the same condition.

diode and the PLCM, and ranges theoretically from 0 ms up to a maximum of one cycle time (3 ms). The remaining delay time from the PLC to the MDBRAM and from the MDBRAM to the PLCM is mainly caused by sending and receiving data through the interfaces (Pmods, IP-Cores) of the FPGA board as well as the finite data processing speed of the data diode and the modified driver running on the PLCM.

VII. CONCLUSIONS AND FUTURE WORKS

In this study, a data diode built on an FPGA board has been presented. When utilizing available communication interfaces of the PLC, the need of additional hardware for the data diode is minimized and the integration of the data diode into the PLC system can be achieved with a minimum of efforts for the user. A well-known Open-PLC product line called RevPi is selected to test the implemented data diode. By sniffing the data on the backplane bus of the RevPi controller within the OT system, the data diode is able to collect data without affecting the architecture of the protected system and its normal process operation. Employing the MDBRAM, data is unidirectionally transferred from the high security zone to the low security zone. In addition, two software modes based on different criteria have been developed for the data diode to communicate with another controller called PLCM on the IT side. In the first mode of operation the data diode emulates a regular gateway module, which makes the implementation of the data diode straightforward. In the second mode, the data diode operates with the modified driver of the RevPi backplane bus running on the PLCM to reflect the entire protected PLC system behind the high security zone. The second mode solves the disadvantages of the first mode such as length limitation of the transmitted data and the change of the architecture from the point of view of the control system (e.g. SCADA). However, the user must install the modified driver for the PLCM before the second mode can be used. In the rest of the paper, the performance of the data diode has been tested with a dedicated test bench.

The approach presented here can be applied to any PLC if the communication protocol running on the backplane bus is known. In further developments improvements will be made, e.g. also including security features such as authentication of the data diode in the OT and IT network.

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